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A BENCHMARK FOR QUANTUM OPTIMIZATION: THE TRAVELING SALESMAN PROBLEM

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We present compelling reasons for symmetric traveling salesman problems (TSPs) to be the benchmark for quantum computing of combinatorial optimization problems for all types of quantum hardware. There are seven reasons for endorsing these TSPs to be the benchmark and no shortcomings.

Keywords: quantum computing, combinatorial optimization, exact solution, speedup, traveling salesman problem, TSP library

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1 Introduction

Quantum computing is beginning to break through technical difficulties and emerge as a viable means of computing. As part of its step forward, benchmarks are needed to compare the accuracy and speed of quantum results with each other and with the best classical outcomes. Also, benchmarks will be a means to uniformly rate quantum software and hardware produced by various companies. This paper discusses why the symmetric traveling salesman problem (TSP) is an outstanding benchmark for quantum computing of combinatorial optimization problems.

We do not compare the TSP with other benchmarks in the literature [1] since the TSP is easy to grasp, is implemented on D-Wave hardware, has many applications, and seems to have very little bias. The following benchmarks have been tried on a D-Wave quantum computer and abandoned: quadratic assignment problems with constraints, 2-satisfiability problems, and linear assignment problems [2].

Quantum computing seems to be analogue. Thus, it may produce approximate answers in its present state of development. The classical work in the TSP Library has exact global optimums which is an advantage in Section 3.7. Therefore, we do not address benchmarks for TSP heuristics but only for algorithms designed to find exact answers. This decision possibly eliminates the quantum approximate optimization algorithm (QAOA) as a benchmark. Based on the scaling of circuit depth, it is argued in [3] that the TSP is not suitable for QAOA. On the other hand, QAOA seems to be the standard algorithm for solving some combinatorial optimization problems on a quantum gate simulator [4].

Paper [5] addresses factors that determine the performance of optimization problems on quantum annealing devices. A supplication contains a theoretical discussion of implementing them on the TSP. Computational information is not included, leading to the conclusion that no TSPs were solved.

Section 2 of this paper has a brief description of the traveling salesman problem. Section 3 contains seven reasons supporting the symmetric TSP as a benchmark for quantum combinatorial optimization. Section 4 describes two possible drawbacks for asymmetric TSPs to be a benchmark. Conclusions are in Section 5. An Appendix contains benchmarking principles.

2 Description of the traveling salesman problem

The traveling salesman problem (TSP) is defined by a list of cities, including the salesman's home city, and the distance between each pair of cities. The distances are undirected. The TSP asks for a shortest route for the salesman to visit each city once and return to his home city. This is called the symmetric TSP. If the distances are directed so that the distance from city A to city B may be different than the distance from city B to city A, then the TSP is called asymmetric.

Equivalent formulations can be found in symbols [6, page 1], linear programming [7, pages 1-2] and graph theory [8, page 6].

3 Seven reasons to use the symmetric TSP as a benchmark for quantum combinatorial optimization problems

In this section we briefly explore the advantages of the symmetric TSP to act as a benchmark for quantum optimization problems. The list of advantages approach benchmarking differently than the full list of benchmark principles in the Appendix. There is no structural match between the two lists.

3.1. Renowned, Recognized Problem

The TSP is a well-known, combinatorial optimization problem. This is shown by reference books [7-12] and a countless number of papers dating back to at least 1954 [13].

3.2. Library of TSP Examples

The TSP library (TSPLIB) [14] is a list of accessible problems that have become the standard test instances for TSPs. This means all researchers can use the same TPSs which facilitates comparisons. The problems in the TSPLIB have been solved, i.e., the length of a shortest route has been found and published in [14] but apparently a shortest route has not been published.

The TSPLIB has 110 symmetric TSPs that range in size from 14 cities to 85,900 cities. The library is stable because it has been closed to new problems since 1/1/2013. Additional challenging TSPs are listed on the website [15].

3.3. Well Established in Quantum Annealing

The TSP has been studied in the quantum annealing framework. The TSP is well formulated for quantum annealing in [16-17] which have the same formulation from different approaches. Details of the formulation are described in Python code in a D-Wave Jupyter Notebook for a 48-city TSP. The formulation has the advantage of preventing subloops, which means they do not need to be blocked by constraints.

D-Wave provides a 7-city TSP and a 48-city TSP, both symmetric, in its LEAP tool kit as examples for quantum annealing. The TSP code is shown and can be executed. An exact solution is not given. (A LEAP logon ID and password are needed. They are provided when signing up for one minute of free computing time on a D-Wave processor.)

Small instances from the TSPLIB have been studied for accuracy using quantum annealing. Reference [18] contains a report of a study of four symmetric TSPs. Likewise [19] using different hybrid solvers reports on studies of four symmetric TSPs and two asymmetric TSPs from the TSPLIB.

The equivalence in paper [20] implies that the gate model for quantum computing has a formulation for the TSP that automatically prevents subloops. However, the theoretical nature of the equivalence makes the equivalence unrealistic to implement. The TSP on a quantum gate chip is in its infant stage. This work began in [21] on a 4-city, symmetric TSP. A 4-city TSP is moderately trivial because subtours do not need to be blocked. None of the citations of [21] solve a larger TSP on a quantum gate chip.

3.4. Computational Complexity

The computational complexity of the TSP is NP-complete [7, Ch. 3], [11, Ch. 9] and [22]. Thus, the TSP is equivalent to more than a hundred problems in the NP-complete class in the sense that a polynomial time solution for one member of the class transfers to all other members.

3.5. Classical Solution Algorithm

The classical software Concorde [15] rapidly solves symmetric TSPs exactly. Concorde seems to have become the gold standard for classically finding exact solutions for symmetric TSPs. Quantum hardware and software are challenged to solve symmetric TSPs exactly and show speedup over the processing time used by Concorde.

3.6. Real Applications

The TSP has many practical applications. It is much more than a theory or a game. See [7, Ch. 2 Sec. 2], [8, Ch. 11], [9, Ch. 2], [10, Ch. 1], and [11, Ch. 3].

3.7. Precise Solutions, Not Approximations

Exact global optimum, which is the length of a shortest route for the salesman, is published in [14] for all TSPs in the TSP Library.

All instances of the TSP that we know about, and were processed on the chip of a D-Wave quantum machine, and used the formulation in [16-17] have returned exact solutions. For example, [23] reports exact solutions for four 6-city TSPs.

When the TSP was too large to be processed on the chip, [18] reports use of the D-Wave tool qbsolv which is a classical solver that can also implement a quantum annealer for a hybrid mode. It is based on tabu search. Qbsolv combines results for an answer for the TSP. According to Table 1 in [18], this method gave an exact answer for 100% of the 100 executions to solve the symmetric 14-city TSP Burma14 in the TSBLIB.

4 Difficulties for asymmetric TSPs to be benchmarks

In this section we discuss two difficulties that asymmetric TSPs have.

4.1. No Classical, Universal Solution Algorithm

Apparently, there is no classical, gold standard for rapidly finding exact solutions for asymmetric TSPs. Thus, there seems to be no classical baseline to compare quantum results for these TSPs. Moreover, there has been no published work about solving the asymmetric TSPs in the TSPLIB, which could be a challenging open problem.

As recently as 2019 an improved algorithm was proposed for classically solving asymmetric TSPs [24], but not demonstrated on TSBLIB problems. Previous work on this topic has been published in [10, Ch. 4] and [25-27].

4.2. Large Number of Connections Between Qubits

If N is the number of cities in a TSP, then the quantum formulation of an asymmetric TSP has $(N - 1)(N - 2)^2 + 2(N - 1)$ connectivity between qubits. The large number of connections makes the asymmetric TSP difficult to solve by quantum annealing based on [16] and [17]. The corresponding formula for a symmetric TSP is $(N - 1)(N - 2) + (N - 1)$. Comparing the two formulae, the asymmetric is larger by a factor of O(N).

5 Conclusion

Everything seems to be in place for classical, symmetric TSPs to be a benchmark for quantum computing of combinatorial optimization problems. Now we are waiting for quantum hardware to catch up to be able to execute symmetric problems in the TSP library without hybrid methods.

The answer is not as clear for classical, asymmetric TSPs to be a benchmark for quantum computing of combinatorial optimization problems. This is due to no standard, classical software for solving asymmetric TSPs and a very large number of qubit connections between cities and locations on a route.

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Appendix: Benchmarking Principles

According to [28], the seven benchmarking principles are:
Relevance: Benchmarks should measure relatively vital features.
Representativeness: Benchmark performance metrics should be broadly accepted by industry and academia.
Equity: All systems should be fairly compared.
Repeatability: Benchmark results can be verified.
Cost-effectiveness: Benchmark tests are economical.
Scalability: Benchmark tests should work across systems possessing a range of resources from low to high.
Transparency: Benchmark metrics should be easy to understand.